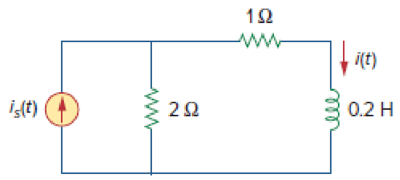
Chapter 16, Problem 14P

Problem

Find *i*(*t*) for *t* > 0 for the circuit in Fig. 16.37. Assume *is*(*t*) = [6(*t*) + 3δ(*t*)]mA.

Figure 16.37:



Step-by-step solution

Step 1 of 6

Refer to Figure 16.37 in the textbook.

There is typo error in the expression of source current. The expression for the source current is,

$$i_s(t) = [6u(t) + 3\delta(t)] \text{ mA}$$

Apply Laplace transform on both sides.

$$I_s(s) = \frac{6\times10^{-3}}{s} + 3\times10^{-3}$$

The inductance in s-domain is,

$$\begin{aligned} Z_L &= sL \\ &= 0.2s\ \Omega \end{aligned}$$

Step 2 of 6

Draw the circuit in s-domain.

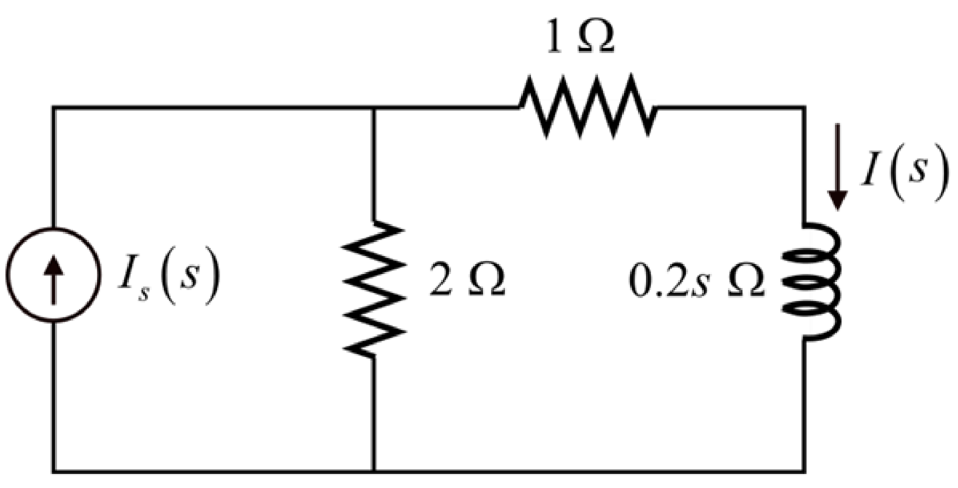


Figure 1

Step 3 of 6

Apply current division rule to calculate the expression for $I(s)$.

$$I(s)=\left(\frac{2}{2+1+0.2s}\right)I_s(s)$$
$$=\left(\frac{2}{3+0.2s}\right)\left(\frac{6\times10^{-3}}{s}+3\times10^{-3}\right)$$
$$=\left(10^{-3}\right)\left(\frac{10}{s+15}\right)\left(\frac{6}{s}+3\right)$$
$$=\left(10^{-3}\right)\left(\frac{10}{s+15}\right)\left(\frac{6+3s}{s}\right)$$
$$=\left(10^{-3}\right)\left(\frac{60+30s}{s(s+15)}\right)$$

Step 4 of 6

Apply partial fractions to the function $\frac{60+30s}{s(s+15)}$.

$$\frac{60+30s}{s(s+15)}=\frac{A}{s}+\frac{B}{s+15}$$

Calculate the value of the unknown, A.

$$A=s\left[\frac{60+30s}{s(s+15)}\right]\Bigg|_{s=0}$$
$$=\frac{60+30s}{(s+15)}\Bigg|_{s=0}$$
$$=\frac{60+0}{0+15}$$
$$=4$$

Step 5 of 6

Calculate the value of the unknown, B.

$$B=(s+15)\left[\frac{60+30s}{s(s+15)}\right]\Bigg|_{s=-15}$$
$$=\frac{60+30s}{s}\Bigg|_{s=-15}$$
$$=\frac{60+30(-15)}{-15}$$
$$=26$$

Step 6 of 6

The expression for $I(s)$ is,

$$\begin{aligned}
I(s) &= \left(10^{-3}\right)\left(\frac{60+30s}{s(s+15)}\right) \\
&= \left(10^{-3}\right)\left(\frac{4}{s}+\frac{26}{s+15}\right) \\
&= \frac{4\times10^{-3}}{s}+\frac{26\times10^{-3}}{s+15}
\end{aligned}$$

Apply inverse Laplace transforms on both sides.

$$\begin{aligned}
i(t) &= \left(4\times10^{-3}\right)u(t)+\left(26\times10^{-3}\right)e^{-15t}u(t) \\
&= \left(4+26e^{-15t}\right)u(t)\ \text{mA}
\end{aligned}$$

Thus, the expression for the output current is $\boxed{\left(4+26e^{-15t}\right)u(t)\ \text{mA}}$.